

Text2AC: A Framework for Game-Ready 2D Agent Character(AC) Generation from Natural Language

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Figure 1: Different characters created by users through Text2AC

ABSTRACT

The visual appearance and personality of game characters are crucial to the player experience. Assembling characters in 2D games is a common practice with a significant demand for creativity, yet the creation process remains challenging, confined to professional graphic designers and animators. Traditional automated character creation in games usually involves manual parameter adjustment or the use of photos as references, coupled with the selection of pre-set backgrounds or professions within the game, leading to limitations in terms of freedom, usability, and consistency. However, with advancements in natural language processing, large language models (LLMs), and generative AI, we have the opportunity to reimagine the process of user-interactive character creation. This paper introduces a novel text-to-character methodology, integrating LLM technology with generative image AI, eliminating the need for reference photos or manual parameter editing while providing extensive options for character creation. Through natural language processing and automated skeletal animation, the generated 2D characters are capable of being driven across various web platforms

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and game engines. This exploration aims to establish a framework for driven agents that harmonizes the visual layer with linguistic personality, thereby providing a more enriched user experience in games and other applications as agents.

CCS CONCEPTS

• **Human-centered computing** → **Interactive systems and tools.**

KEYWORDS

Game Character Generation, Generative AI, Personalized Agents, 2D Character Animation

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1 INTRODUCTION

In gaming, personalized character creation, reflecting stronger individual player traits and attributes, is increasingly becoming a focal point for players. Studies suggest that the alignment of a player's ideal self-image with their character positively impacts player-character interaction and in-game perception [6]. Thus, for

role-playing games, the character creation process is a crucial component [13], demanding further enhancement in character appearance, personality, and actions.

Current mainstream games typically establish character appearance and personality by offering a range of appearance parameters, allowing for the adjustment of physical and facial features, and defining character personality and skills through the selection of various in-game racial and professional systems (e.g., *The Elder Scrolls* series). This approach is often restricted by fixed templates and complex parameters, presenting challenges for non-artist players in creating characters that match their initial expectations [14]. Existing technologies have not yet provided a comprehensive and efficient solution for the user-driven generation of game characters, creation of animation assets, and personalized dialogues [12]. Researchers are exploring new tools and methods, like deep learning and image style transfer, for game character image generation [12, 14], but these are generally suited for 3D workflows and heavily rely on players' real photo inputs, limiting freedom in fantasy character styles and attire.

Games are increasingly seen as a significant field of study in human-agent interaction [19], where the rapid development of Artificial Intelligence (AI) technologies has led to tools capable of generating various styles of 2D images. Furthermore, the emergence of Large Language Model (LLM) technology has created new possibilities for personalized dialogue creation in game characters [1]. Agents, as intelligent entities, can enhance the interaction experience within an application interface through the presentation of a character image. Such game assets, which can serve as in-game characters or agents in games and other applications, are referred to in this paper as agent characters (AC).

2D personalized game character assets are crucial in game design, known for their freedom, complex mesh deformation, and artistic quality [8]. However, their creation predominantly depends on skilled game artists and animators using tools like Spine or Live2D for assembling layered designs into game characters, a process entailing substantial specialized manual work. This reliance on expert skills poses a notable challenge to automating 2D character generation. Therefore, we explore how AI techniques can optimize the character creation process.

In this paper, we introduce Text2AC, a text-based agent character (AC) creation system that rapidly constructs character images, personalities, backgrounds, and skills aligned with highly flexible user descriptions via natural language input. Utilizing a universal character template, the system performs image extraction and skeletal binding. The AC assets, drivable by programs or agents, can interact with users through animation and text in various games or applications. To assess the system's effectiveness, we conducted a user study with 8 participants, comparing Text2AC with existing character creation processes. Participants found Text2AC to be more straightforward in creating characters that met their expectations and offered greater freedom. Overall, our work contributes the following:

- (1) The design and implementation of Text2AC as a comprehensive character creation system, encompassing character appearance and personality. It enables ordinary players to use natural language to generate diverse characters which

can be deployed across different web platforms and game engines as an agent driver.

- (2) We propose a method for translating user natural language into image prompts and creating character backgrounds and personalities. This approach, leveraging natural language inputs and randomness in prompts, aids players in crafting detailed character descriptions.
- (3) The conversion of text-generated images into 2D virtual characters capable of action and behavior across multiple platforms. To our knowledge, this is among the earliest research to use generative AI imagery for creating drivable agent characters.
- (4) A user study with 8 participants demonstrating the advantages of the Text2AC system in character creation tasks.

2 RELATED WORK

2.1 The automated creation of game characters

In the context of establishing the relationship between players and characters within a game, the character creation process is a key component [6]. Games provide players the opportunity to express their identity and cultivate abilities through the creation and development of their characters. Players tend to perceive their virtual selves as an alternate form of ideal self-presentation in the virtual world [5, 9]. Player identification with their created game characters often leads to an affinity for the game [4]. Numerous studies aim to optimize the experience of character creation for players, enhancing freedom and reducing complexity. Procedural Content Generation (PCG) is a commonly employed technique in game character generation, typically utilizing modifiable specific features to enhance the personalization of character appearance, changeable within the framework of game rules [3, 11]. For instance, Wolf et al. employed Tied Output Synthesis (TOS) to predict a set of binary facial parameters, controlling the graphical engine to generate a character resembling the human in an input photo [16]. Similarly, Shi et al. used methods like Face-to-Parameter (F2P) and style transfer to establish characters matching the input photo's appearance [10]. However, these methods require reference facial photos for automatic creation. In contrast, textual prompts offer a more flexible and time-saving approach for the automatic creation of game characters. Zhao et al. proposed a novel Text-to-Parameter Translation method (T2P) for text-driven creation of 3D game character facial parameters [18]. These works are primarily focused on 3D processes and are limited to the automated alteration of character appearance within a fixed model, unable to automatically create an overall character style, attire, or even personality. Our goal, however, is to establish a fully drivable in-game agent character based on textual input, simultaneously creating a matching background and personality.

2.2 Text-based content generation tools

The Text-to-Image generation model refers to a category of machine learning models that generate images matching a given language description as input [7]. Stable Diffusion employs Latent Diffusion Models to generate images and is open-sourced to the public, gaining widespread popularity and usability [1, 2]. The advancements in Large Language Models (LLM) and generative AI technologies offer



Figure 2: User interface

new perspectives or solutions in various domains of content production and interaction. Yan et al. proposed a content production tool utilizing technologies like GPT-3 and Dalle-2, merging text and image inputs to assist in creating narrative works[17]. They employed SpaCy to split text inputs into clauses, utilized GPT-3 to validate the processed tuples <subject, predicate, object>, and provided text completion. Ashby et al. leveraged LLM technology and user input to generate game tasks and create fluent, unique NPC dialogues based on the game state[1]. Additionally, there are several works optimizing text-based generation experiences. Wang et al. developed RePrompt, an automated method refining text prompts to optimize image generation for precise emotion expression[15]. Brade et al. created Promptify to support the text-to-image workflow, encompassing prompt writing, image generation and review, and refining initial prompts[2].

3 USER INTERFACE AND INTERACTION

To verify the feasibility of creating a text-based agent character, we developed a web-based interactive character creation interface. Users begin by entering a natural language description of the character's appearance (Figure 2a), with the option to detail clothing styles, facial expressions, clothing types, or even eyelash colors, or simply specify gender. For random character creation, users can click the "Random" button, prompting the system to generate a comprehensive appearance description. Upon finalizing their description and clicking "Generate," the system swiftly produces a character illustration and a dynamically drivable AC.

Further, users can describe the AC's personality, background, and occupational skills in natural language. The system seamlessly

integrates these inputs into the character panel (Figure 2c). Users can then view and examine their created characters (Figure 2b), including the AC (Figure 2d), which is universally drivable across platforms like web, Unity, and various game engines. Powered by LLM, this agent character enables user interaction and development through natural language.

4 SYSTEM OVERVIEW

Text2AC introduces a two-step system for the automated creation of agents, involving the generation of 2D appearance and core characteristics. The system streamlines the workflow through four key components: (1) Text2Text, which converts appearance text descriptions into crafted prompts; (2) Text2Role, translating crafted text prompts into game characters which can be driven as an agent; (3) Text&Image2Image, combining crafted prompts and game characters images to create consistent characters (tachie appearances); and (4) Text2Text, transforming background descriptions into agent characteristics. By integrating these core components, Text2AC implements an efficient and user-friendly workflow, enabling users to generate a complete agent role with unique appearance and characteristics. Figure 3 illustrates the entire Text2AC process. The subsequent subsections provide insights into the design and features of each component within these two steps.

4.1 Appearance Generation

The first step for user is to create a 2D outlook and portable game character with only inputting the desire text descriptions in a natural language manner. This step eliminates the tedious steps in

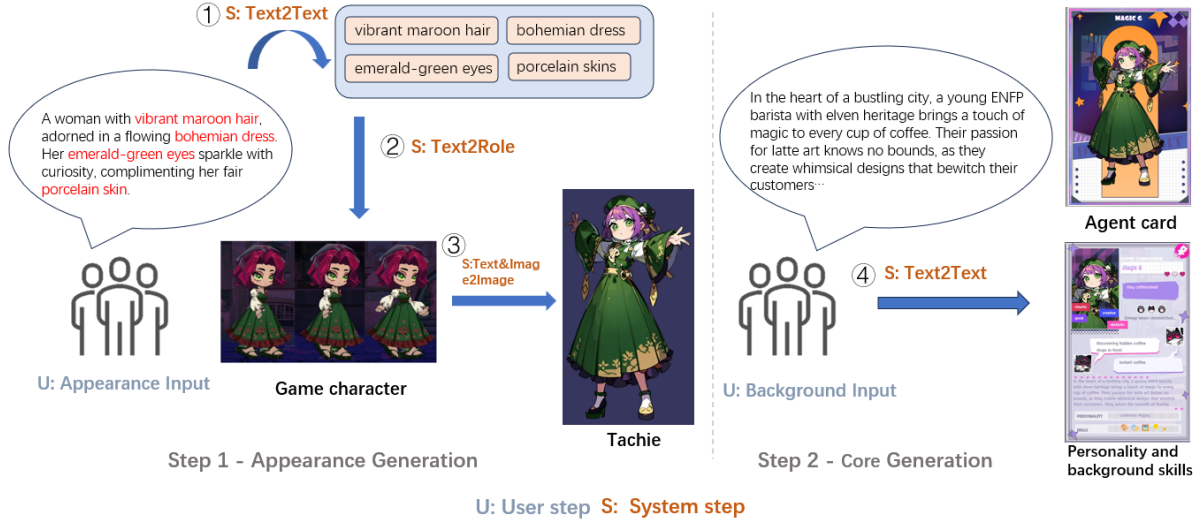


Figure 3: Basic process for creating a agent character(AC): Appearance&Core

the traditional character creation process with automatically implementing three components.

Text2Text: appearance text description to crafted prompt. Although deep learning models e.g. StableDiffusion and DALL-E provides powerful image generation ability with text input, writing effective prompts to creating a expected character based on these AI models is still challenging for users [2]. To tackle this problem, we implement text-to-text mapping system to automatically transfer the user input into crafted prompt for later image generation. Hence, users are allowed to input agent outlook in natural language descriptions. As show in the Figure 3, key informations about character's outlook will be extracted into crafted prompts for the next step. In addition, the system provides a user-friendly creation process via giving automatic outlook prompt recommendation with using large language models (LLMs) e.g. ChatGPT. Users can easily edit the generated outlook example or reroll a new one in the system.

Text2Role: crafted prompt to game character. In this step we hope to generate a game character that can be driven as an agent, including appearance and controllable skeletal deformation animation. our goal is to establish an automated, universal creation process that can be applied seamlessly across different web platforms and game engines, facilitating the migration and application of characters.

Taking humanoid characters as an example, we first define a uniformly proportioned character template, as shown in the figure. Based on this template, we collect and create a template library that includes outline images of different clothing and hairstyle types. These images are categorized based on their attributes, facilitating rendering sequence determination after differentiation. After the user inputs a character description through natural language, we extract the crafted prompt through the above process. With this information, we use the template library to synthesize an outline drawing, serving as a reference for Body generation, while also

recording the component information that constitutes the outline drawing for reference in the subsequent segmentation.

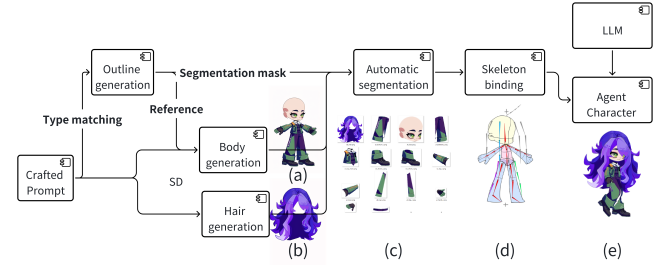


Figure 4: Workflow from crafted prompt to a game character that can be driven by LLM

By combining the prompt with the outline as input for Stable Diffusion (sd), we generate personalized character bodies and hairstyles that conform to our uniform character proportions (Figure 4a,b). We train a model based on Low-Rank Adaptation (LoRA) to stabilize the style and posture of the generated images. Subsequently, based on the previous outline mask information and the Segment model, we automatically clip the character into different images based on body parts' hierarchy (Figure 4c) and perform skeleton binding using a 2D animation software, Spine (Figure 4d), driving smooth deformation animations with elastic bones and Free-Form Deformation.

To increase the freedom in generating clothing and automate the binding, we divide the character's body into different regions of meshes to store images of different types (Figure 5a), and unify their binding and deformation weights in Spine. Thus, for clothing of the same type, different sizes and styles can be freely generated. For example, with a universal deformation mesh, it is possible to

automatically deform and bind various types of shoes, from sandals to boots.



Figure 5: (a)Under the universal skeleton, different types of clothing automatically use different meshes, and the same type of clothing in various styles can be stored in the same mesh. (b)Generate consistent game characters and tachie through image interrogate text and image input

Text&Image2Image: consistent character tachie generation.

Based on the generated appearance of the game character, we aim to produce a character illustration with different artistic styles for static display. This requirement is common in the gaming industry, and the resulting illustration is referred to as the character's tachie. As shown in Figure 5b, we present two automatically generated game characters along with their tachie. In this step, we employ a carefully trained style transfer SD model to control the artistic style of the generated tachie. Using the game character images as a reference and employing an image interrogation model as a prompt, we generate static images that are highly consistent with the game characters but possess a stable tachie artistic style.

4.2 Core Generation

The second step for user is to create a role card with the background story and key characteristics of an character. The system optimizes the creation process with automatically transferring the background story to characteristics of the character.

Text2Text: background description to character characteristics. Different from traditional characters having fixed attributes, the system provides extra flexibility in agent creation by allowing unlimited background story options and characteristics descriptions for each agent. These helps user to assign a unique core for each agent which is the key information to perform uniquely in various application scenarios. Similar with the first step, the system provides a random background recommendation via large language models (LLMs). User can also edit the background information or generate new descriptions randomly in the system. In addition, we design a specific prompt template to concatenate with background description as the input of LLMs. With the prompt, LLMs extract the key information in the background and generate the key characteristics of a agent e.g. like, dislike, skills, and personality etc(Figure 2c). In order to provide maximum customization ability in agent creation, user is capable of editing each attributes of the character in the edition page.

5 USER INTERVIEWS AND DISCUSSIONS

We invited 8 gamers with over 10 years of gaming experience on average and prior experience in creating game characters in role-playing games to participate in user interviews for this research. These participants were first asked to thoroughly experience the entire character creation workflow along with two key application modules - chat and sandbox. We then conducted a series of inquiries regarding their satisfaction levels on the efficiency and performance of character creation as well as the experiences of interacting with the generated characters.

In terms of efficiency and performance, these 8 participants spent to complete the entire character creation process was 3 minutes on average. All participants agreed that Text2AC's creation workflow was markedly more efficient compared to previous approaches, as they could simply produce expected character appearances from language descriptions rather than tedious facial parameter tuning. Additionally, 75% of participants were satisfied with the visual quality and intricacy of the generated characters. Over half of them specifically complimented the high diversity encompassing facets like visages, hairstyles, outfits, and styles. The remaining 25% felt some particular physical trait descriptions were not accurately captured and reflected in the final renderings. As for the translation of character backgrounds and personalities from texts, 87.5% considered the outcomes aligned with their inputs. They remarked that the system excelled in comprehending the provided information and extracting key semantic elements to constitute personalized character parameters.

Overall, Text2AC offers efficient means for automated character assembly utilizing language depictions. By significantly streamlining the workflow, visuals and attributes are synthesized to largely meet user expectations. There was widespread approbation for the proposed creation approach and subsequent applications. Our future efforts will concentrate on improving text-image matching precision and strengthening cross-module coherence to achieve richer, more concrete and satisfying experiences.

6 LIMITATIONS AND FUTURE WORK

Text2AC demonstrates potential in automating 2D agent creation but faces challenges that future research could address to enhance its effectiveness. A key limitation is the difficulty in handling complex or unusual user requests and intricate backgrounds, a result of the models being trained on limited datasets. Enhancing algorithms with more diverse and rare data could bolster the system's robustness and expand its application range.

The impact of personalized roles on user experience and game interaction also requires further investigation. Future developments should refine the generation process, incorporating adjustable parameters for detailed customization and an interactive user interface. Comprehensive user studies across various gamer demographics would be instrumental in optimizing the system for enhanced enjoyment and engagement.

Another challenge is managing the uncertainty of AI outputs and the complexity of asset creation in interactive gaming systems, particularly balancing asset flexibility with gaming suitability. We suggest translating user descriptions into templates matching the generative engine's input for improved consistency and implementing a unified character rig system for diverse styles. Additionally, preset world views and examples could guide AI-generated text for better in-game applicability.

The potential to extend this technology beyond gaming into social platforms and virtual spaces is particularly intriguing. Automated translation from text to visual profiles could revolutionize identity expression in online communities. Moreover, immersive 3D environments with text-driven avatars promise more personalized and vivid experiences. This technology could find applications in existing 2D MMO games, where players could design unique characters, enhancing their immersion and engagement. In agent sandbox environments, it could enable users to create and interact with customized agents, opening new avenues for research in interaction design and user psychology. Additionally, the process could be utilized to develop specialized agent assistants, providing personalized support and interaction in various contexts. Such applications would not only broaden the scope of personalized character customization but also offer valuable insights into user preferences and behavior in digital spaces.

7 CONCLUSION

In this paper, we present a text-based character creation process aimed at exploring the application of generative AI and Large Language Model (LLM) technologies in the realm of game interaction, offering a simpler and more diverse character creation experience. We developed an algorithm for translating natural language into prompts, facilitating the generation of character appearance and background. Additionally, utilizing the SD Lora model and auto segment algorithm, we segmented and assembled static character images onto dynamic skeletons. Furthermore, we established a web-based character creation interface, enhancing users' experiences in customizing characters. Through a straightforward user experiment and interviews, we demonstrated the process's ease of use and the system's diversity, providing most users with a novel experience. We also explore how this process opens new avenues for future game character and agent creation, along with potential

research directions, such as investigating the impact of personalized virtual avatars or game characters on gaming experience and social interaction in the human-computer interaction field.

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